

Pendulum $ml^2\ddot{\Theta} = -mgl\sin\Theta - b(\dot{\Theta}) + \tau_m$

Linearization has

$$E = \begin{bmatrix} 1 & 0 \\ 0 & ml^2 \end{bmatrix} \quad F = \begin{bmatrix} 0 & -1 \\ mgl\cos\Theta^*(t) & b'(\dot{\Theta}^*(t)) \end{bmatrix}$$

$$G = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

and

$$A(t) = -E^{-1}F = \begin{bmatrix} 0 & 1 \\ -\alpha_0(t) & -\alpha_1(t) \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ \beta_0 \end{bmatrix}$$

$= -E^{-1}G$

where

$$\alpha_0(t) = g/l \cos\Theta^*(t)$$

$$\alpha_1(t) = \frac{b'(\dot{\Theta}^*(t))}{ml^2}$$

$$\beta_0 = 1/ml^2$$

Suppose output $y = \Theta \quad [z_1(t)]$

$$\Rightarrow y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_C \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_D u$$

Equivalent I/O DE for linearized

output $y(t) = \Theta(t) - \Theta^*(t)$

satisfies

$$\ddot{y}(t) + \alpha_1(t)\dot{y}(t) + \alpha_0(t)y(t) = \beta_0 u(t)$$

= Do properties of this DE change for Diff. ref sol'n's? If so how?

= Consider the equilibria sol'n's for which $\ddot{z}^* = \underline{0} \Rightarrow \underline{z}^* = \text{const.}$ for such sol'n's

$$\alpha_0(t) = g/l \cos\Theta^* \quad (\text{const})$$

$$\alpha_1(t) = \frac{b'(0)}{ml^2} \quad (\text{const})$$

So Dynamics are LTI.

Suppose for convenience numerical

values $g/l = 5, \quad \frac{b'(0)}{ml^2} = 2$

$$\#1) \quad \Theta^* = 0 \Leftrightarrow \mu^* = 0$$

(equil cond'n $mgl\sin\Theta^* = \mu^*$)

char poly is $s^2 + 2s + 5$ / poles $\begin{matrix} -1+2j \\ -1-2j \end{matrix}$

$$\Rightarrow y_h(t) = C_1 e^{-t} \cos(2t) + C_2 e^{-t} \sin(2t) \\ = A e^{-t} \cos(2t + \varphi)$$

$$\text{Stable: } y_h(t) \rightarrow 0$$

$$\Rightarrow \Theta(t) \rightarrow 0$$

$$2.) \Theta^* = \pi/4 \Rightarrow \alpha_0 = \frac{g}{\sqrt{2}} \approx 3.54$$

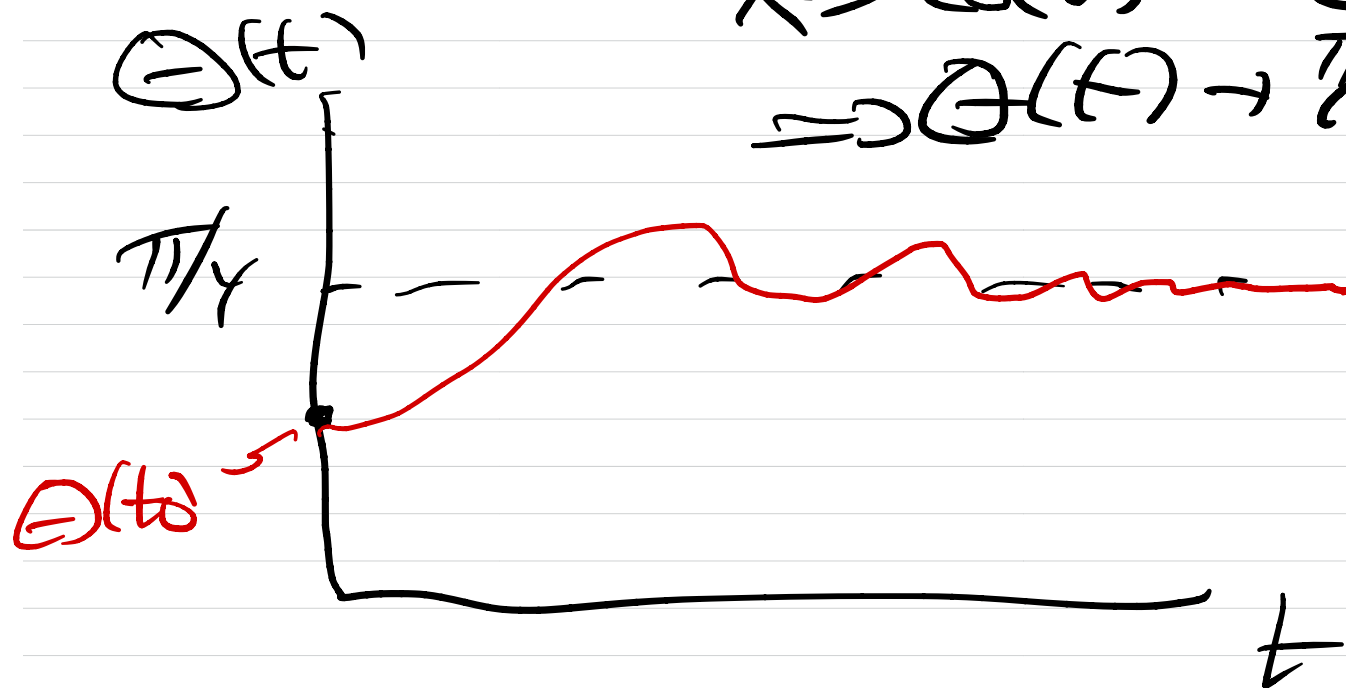
$$\Rightarrow \text{poles } -1 \pm 1.59j \quad \text{freq changed}$$

$$\Rightarrow y_h(t) = A e^{-t} \cos(1.59t + \varphi)$$

$y_h(t) \rightarrow 0$ stable.

$$\Rightarrow \Theta(t) \rightarrow 0$$

$$\Rightarrow \Theta(t) \rightarrow \pi/4$$



$$\#3 \quad \Theta^* = 80^\circ$$

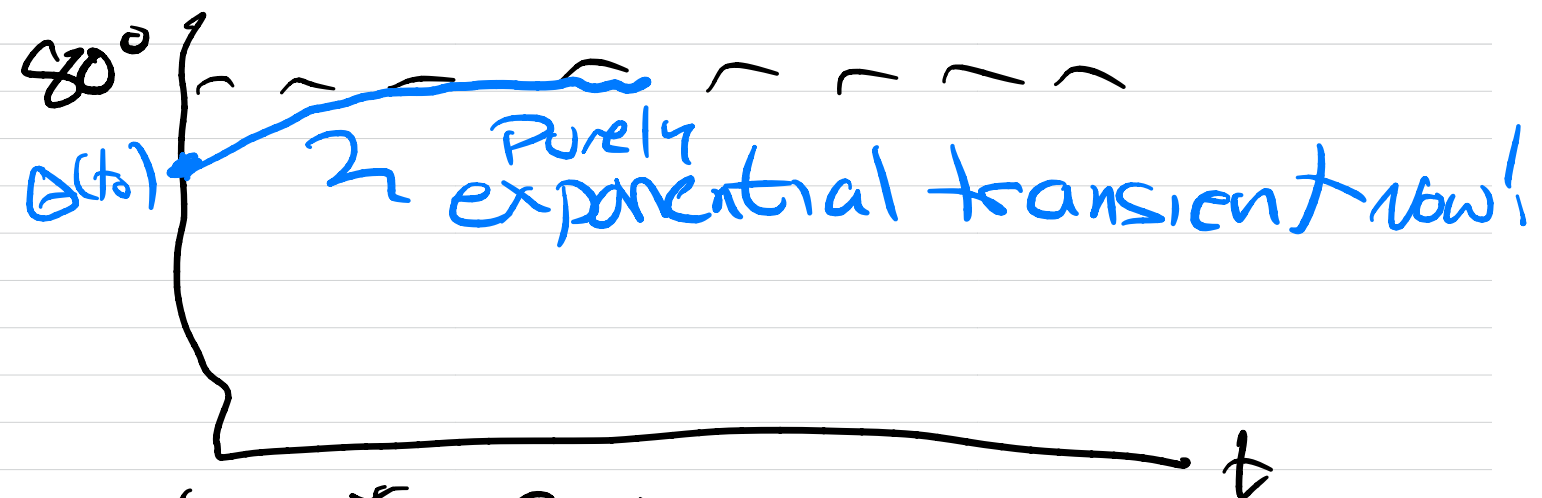
$$\alpha_0 = g/l \cos 80^\circ \approx 0.868$$

$$\text{Char poly } s^2 + 2s + 0.868$$

$$\text{has poles } -1.36, -0.64$$

$$\Rightarrow y_h(t) = C_1 e^{-1.36t} + C_2 e^{-0.64t}$$

$$\text{Stable} \Rightarrow y_h(t) \rightarrow 0$$



$$\#4 \quad \Theta^* = 90^\circ \Rightarrow \omega^* = \sqrt{mg/l}$$

$$\alpha_0 = g/l \cos 90^\circ = 0$$

$$\text{Char poly } s^2 + 2s$$

$$\Rightarrow \text{poles } 0, -2$$

$$\Rightarrow y_h(t) = C_1 + C_2 e^{-2t}$$

$y_h(t) \not\rightarrow 0$ here. Not stable

Revisit #1:

Equl condn $mg \ell \sin \Theta^* > \mu^*$

$\mu^* = 0 \Rightarrow \Theta^* = 0$ or $\Theta^* = \pm 180^\circ$

for $\Theta^* = 180^\circ$, $\alpha_0 = g/\ell \cos \Theta^* = -g$

Char poly is $s^2 + 2s - g$

w/ poles $-3.45, 1.45$

and $y_u(t) = c_1 e^{-3.45t} + c_2 e^{1.45t}$

Note $y_u(t) \rightarrow \infty$ here (unstable)

Vector space V , a collection of objects ("vectors") $\underline{v} \in V$ that satisfy

1.) $\forall \underline{v} \in V, \alpha \underline{v} \in V, \alpha \in \mathbb{R}$

2.) $\underline{v}_1, \underline{v}_2 \in V \Rightarrow \underline{v}_1 + \underline{v}_2 \in V$

3.) $\exists \underline{0} \in V \text{ s.t. } \underline{v} + \underline{0} = \underline{v}$

Typically in this class $V = \mathbb{R}^n$ for which $\underline{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$ $\alpha \underline{v} = \begin{bmatrix} \alpha v_1 \\ \vdots \\ \alpha v_n \end{bmatrix}$

$$\underline{v} + \underline{w} = \begin{bmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{bmatrix}$$

[We will sometimes use $V = \mathbb{C}^n$ for which $v_k = \theta_k + j\omega_k$]

Will use " τ " to denote vectors w/horiz orientation if needed $\underline{v}^\tau = [v_1 \dots v_n]$

Dot product: $\underline{v} \cdot \underline{w} = v_1 w_1 + \dots + v_n w_n$
 $= \sum_{k=1}^n v_k w_k$

W is a subspace of V
 $W \subset V$ if W satisfies the vector space axioms in it's own right.

i.e.

$$W = \left\{ \begin{bmatrix} a \\ b \\ 0 \end{bmatrix}, a, b \in \mathbb{R} \right\} \subset \mathbb{R}^3$$

- Linear combination any expression of form

$$\alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \dots + \alpha_p \underline{v}_p$$

$$= \sum_{k=1}^p \alpha_k \underline{v}_k \quad \alpha_k \in \mathbb{R}$$

i.e. a weighted sum of vectors

- Span: The span of a collection $\{\underline{v}_k\}_{k=1}^p$ is the set of all possible linear comb. of the vectors in the collection.

This defines a subspace of V

so for example, above $W \subset \mathbb{R}^3$
 $= \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

- A collection $\{\underline{v}_k\}_{k=1}^p$ is linearly independent iff

$$\sum_{k=1}^p \alpha_k \underline{v}_k = \underline{0}$$

$$\iff \alpha_k = 0 \quad k=1, \dots, p$$

i.e. only possible linear comb with result $\underline{0}$ is the trivial one $\alpha_1 = \alpha_2 = \dots = \alpha_p = 0$

- The dimension of a collection $\{\underline{v}_k\}_{k=1}^p$ is the maximum number of lin. indep vectors in the collection.

• A collection $\{\underline{v}_k\}$ is a basis for V , if

1.) $\text{span}\{\underline{v}_k\} = V$

2.) Collection $\{\underline{v}_k\}$ lin. indep

$\Leftrightarrow$ Every $\underline{v} \in V$ can be written as a UNIQUE linear comb. of the vectors in the basis.

Note: A vector space can have as many different bases, but all bases will have the same dimension, which defines the dimension of the vector space itself.

for $\mathbb{R}^3$ could use $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$
 or $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$, etc

The 1st of these is the so-called "natural basis". Generically for $\mathbb{R}^n$ this is $\{\underline{e}_k\}_{k=1}^n$,

where

$$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \underline{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \underline{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$$

Natural basis is orthonormal

$$\underline{e}_i \cdot \underline{e}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Matrix: $A_{p \times m}$

$$A = \begin{bmatrix} \boxed{a_{11}} & \boxed{a_{12}} & \dots & \boxed{a_{1m}} \\ \boxed{a_{21}} & \boxed{a_{22}} & \dots & \boxed{a_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \boxed{a_{p1}} & \boxed{a_{p2}} & \dots & \boxed{a_{pm}} \end{bmatrix}$$

$\begin{matrix} \text{row} \\ a_{ij} \in \mathbb{R} \\ i=1, \dots, p \\ j=1, \dots, m \\ \text{col} \\ \underline{a}_j \in \mathbb{R}^p \end{matrix}$

$\underline{a}_1 \quad \underline{a}_2 \quad \dots \quad \underline{a}_m$

So $A = [\underline{a}_1, \underline{a}_2, \dots, \underline{a}_m]$
with $\underline{a}_j \in \mathbb{R}^p$

This is the column partition of A

Matrix-vector mult:

Given $A \text{ } p \times m$, $\underline{x} \in \mathbb{R}^n$

$$\begin{aligned} \underline{y} &= A \underline{x} \\ &= \sum_{k=1}^n x_k \underline{a}_k \end{aligned}$$

Defined only if size of $\underline{x}$ (n)
= # of columns in A (m)

If so, then $\underline{y} \in \mathbb{R}^p$ (p = # rows in A)
and $\underline{y}$ is a linear combination
of the columns of A .